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Intrinsically stretchable quantum dot light-emitting diodes

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Dong Chan Kim[1,2,3,9], Hyojin Seung[1,2,9], Jisu Yoo[4,9], Junhee Kim[1,2,9], Hyeon Hwa Song[5], Ji Su Kim[1,2], Yunho Kim[6], Kyunghoon Lee[5], Changsoon Choi[7], Dongjun Jung[1,2], Chansul Park[1,2], Hyeonjun Heo[1,2], Jiwoong Yang[5] ✉, Taeghwan Hyeon[1,2] ✉, Moon Kee Choi[1,4,6] ✉ & Dae-Hyeong Kim[1,2,8] ✉

Stretchable displays that can change their shape and size under strain could be used to create displays with unconventional form factors. However, intrinsically stretchable light-emitting devices have poor luminous performance, such as low brightness. Here we show that intrinsically stretchable quantum dot light-emitting diodes (QLEDs) can be made using a mechanically soft and stretchable emissive layer consisting of a ternary nanocomposite of colloidal quantum dots, an elastomeric polymer and a charge transport polymer. The light-emitting layer maintains a nearly constant interparticle distance even under 50% strain, ensuring reliable operation of the QLED under stretching. The polymer-rich charge transport region at the bottom of the nanocomposite functions as a hole transport pathway to the embedded quantum dots. The QLEDs exhibit a turn-on voltage of 3.2 V and a maximum luminance of 15,170 cd m^{-2} at 6.2 V without loss of brightness, even when under 50% strain, and can be used to make stretchable full-colour passive-matrix QLED arrays.

Following the development of flexible displays[1–3], researchers have increasingly begun to focus on the creation of stretchable displays[4–6]. There is particular interest in intrinsically stretchable light-emitting devices, including d.c.-driven light-emitting electrochemical cells (LECs) and light-emitting diodes (LEDs) due to their high fill factor, design flexibility and mechanical durability[7–9]. Most reported intrinsically stretchable LECs and LEDs use electroluminescence (EL) of the organic light-emitting composites, which are composed of polymeric light-emitters such as PF-B (a polyfluorene-based copolymer)[10] and SuperYellow (a derivative of poly(1,4-phenylenevinylene))[11], and various additives such as elastomers[12,13], ionic conducting polymer[14] and surfactants[15,16]. In addition to polymeric composites, thermally activated delayed fluorescein emitting polymers with intrinsic elasticity have recently been reported[17]. However, intrinsically stretchable LEDs based on polymeric emitters show limited colour gamut, broad full-width at half-maximum (FWHM) emission spectra and poor luminous performance, such as high turn-on voltage and low brightness (Extended Data Table 1). These issues are primarily attributed to the low quantum efficiency of polymeric light-emitters, as well as inefficient carrier transport to the active layers.

[1]Center for Nanoparticle Research, Institute for Basic Science (IBS), Seoul, Republic of Korea. [2]School of Chemical and Biological Engineering, Institute of Chemical Processes, Seoul National University, Seoul, Republic of Korea. [3]Department of Chemical and Biological Engineering, Gachon University, Seongnam, Republic of Korea. [4]Department of Materials Science and Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, Republic of Korea. [5]Department of Energy Science and Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Republic of Korea. [6]Graduate School of Semiconductor Materials and Devices Engineering, Center for Future Semiconductor Technology (FUST), Ulsan National Institute of Science and Technology (UNIST), Ulsan, Republic of Korea. [7]Center for Opto-Electronic Materials and Devices, Post-silicon Semiconductor Institute, Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea. [8]Department of Materials Science and Engineering, Seoul National University, Seoul, Republic of Korea. [9]These authors contributed equally: Dong Chan Kim, Hyojin Seung, Jisu Yoo, Junhee Kim.
✉e-mail: jiwoongyang@dgist.ac.kr; thyeon@snu.ac.kr; mkchoi@unist.ac.kr; dkim98@snu.ac.kr

A promising alternative to an organic light-emitting layer (EML) is inorganic–organic hybrid EMLs, which consist of inorganic semiconducting nanomaterials and polymeric additives[18–22]. Inorganic nanomaterials, such as colloidal quantum dots (QDs), have been used for the state-of-the-art EL devices, because of their excellent optical characteristics including narrow emission spectra, wide colour gamut and high quantum efficiency[23–26]. However, the development of intrinsically stretchable EMLs based on QD nanocomposites remains challenging. Additionally, similar to purely organic composites, it remains a major technical challenge to facilitate carrier transport to embedded QDs through an insulating matrix. The composition of the intrinsically stretchable EML needs to be carefully optimized to achieve mechanical stretchability and robust charge transport pathways to the QDs for efficient radiative recombination. Also, how the physical and electrical properties of nanocomposites change under applied strain is not well understood, as the spatial organization (such as interparticle distribution) can vary dynamically under mechanical deformations.

In this Article, we report high-performance intrinsically stretchable QD LEDs (QLEDs). The devices have a maximum luminance of 15,170 cd m^{-2} at 6.2 V, a turn-on voltage of ~3.2 V and a stretchability of ~50%. Key to our approach is an intrinsically stretchable EML consisting of a ternary composite of QDs, an elastomer and a charge transport polymer. The addition of an elastomer at a concentration of 10 wt% of the QD weight makes the EML stretchable up to 50% uniaxial strain while maintaining a constant interparticle distance between the embedded QDs. Charge transport polymer-rich islands located near the bottom of the composite provide hole transport pathways to the QDs inside the composite. We use the approach to create full-colour passive-matrix (PM) arrays of intrinsically stretchable QLEDs with transfer-printed red, green and blue (RGB) emissive layers.

QD-based stretchable light-emitting composite

The QD solution (CdSe/ZnS QDs, dispersed in toluene) was mixed with an elastomer solution (polystyrene-block-poly(ethylene-ran-butylene) -block-polystyrene-graft-maleic anhydride (SEBS-g-MA), dispersed in toluene) and then spin-coated on the substrate to form a 40-nm-thick film of the binary QD-elastomer composite (details in Methods). SEBS-g-MA was selected among various elastomer candidates such as poly(styrene–butadiene–styrene) (SBS) or polystyrene-block-pol y(ethylene-ran-butylene)-block-polystyrene (SEBS) because its composite film formed an ultra-smooth surface. As shown in Extended Data Fig. 1a–f, the root-mean-square roughness (R_q) of the thin film of QD/SEBS-g-MA (w/w 10:1) is 1.02 nm, while thin films of QD/SBS or QD/SEBS exhibit higher R_q values (>5 nm) under the same mixing ratio. Moreover, by increasing the contents of SEBS-g-MA to more than w/w 20:1, the roughness of the composite film became even smoother than that of the bare QD film (R_q = 1.57 nm), supporting the homogeneous embedment of QDs inside the SEBS-g-MA matrix.

With increasing weight fraction of SEBS-g-MA, the binary composite of QD/SEBS-g-MA became stretchable (Extended Data Fig. 1g–j). Compared to the QD/SEBS-g-MA (w/w 20:1) film under 50% of stretching, which exhibited partial cracks on the top surface of the films, the QD/SEBS-g-MA (w/w 10:1) film exhibited smoother surface without crack generation under the same stretching condition. The stretchable binary composite film, however, was shown to be unsuitable for stretchable EML of QLEDs, as the elastic polymer greatly inhibited charge transport to embedded QDs. To investigate the carrier transport properties of the binary composite, QLEDs and metal–semiconductor–metal (MSM) devices were fabricated (Extended Data Fig. 1k–n). When we used the binary composite of QD/SEBS-g-MA, the devices showed poor EL characteristics (low brightness and external quantum efficiency (EQE)) and reduced current flows. The current density versus voltage (*J–V*) characteristics of the MSM devices also support the inferior carrier transport properties of the binary composite films. Therefore, additional charge-transporting components should be added to the binary composite to boost charge injection into QDs. Considering the higher degree of the electron injection than that of the hole injection in most QLEDs, which deteriorates EL properties by non-radiative recombination in charged QDs (refs. 27–30), a p-type semiconductor can be a proper choice for the additive to balance the charge injections. Note that the incorporation of n-type semiconducting additives may worsen the charge injection balance, that is, by promoting an excessive injection of electrons.

Figure 1a shows schematic illustrations of the EML for an intrinsically stretchable QLED and its PM array. The stretchable EML is based on a ternary composite of an EL material (CdSe/ZnS QDs), an elastomer (SEBS-g-MA) and a hole transport polymer (poly(9, 9-dioctylfluorene-alt-N-(4-sec-butylphenyl)-diphenylamine) (TFB)). TFB was selected as an additive for the hole transport due to its high hole mobility ($\mu_h \approx 2 \times 10^{-3}$ cm^2 V^{-1} s^{-1}) (ref. 31) and decent solubility in toluene. The composition of the stretchable QD-polymer composite is expressed as $Q_{100}S_xT_y$, which refers to *x* wt% of SEBS-g-MA and *y* wt% of TFB in comparison to the QD weight fraction (100 wt%). A detailed preparation procedure of the composite is described in Methods.

The thin film of the ternary composite exhibits a distinctive internal structure. A cross-sectional transmission electron microscopy (TEM) image depicts the vertical structure of the stretchable EML, exhibiting localized TFB-rich islands at the bottom side of the film (Fig. 1b). These TFB-rich islands cause a large morphological difference between the top and bottom surface (Fig. 1c). While the top surface, which mainly consists of QDs embedded in SEBS-g-MA, has an ultra-smooth morphology (R_q = 0.90 nm), the bottom surface, where aggregated TFB-rich islands are located, exhibits a rough morphology (R_q = 4.30 nm).

Based on our observation of the binary-blended composite films, we inferred the underlying mechanism on the formation of its vertical structure. It reveals that QDs exhibit high miscibility with SEBS-g-MA (Extended Data Fig. 1a–c), whereas TFB does not (Extended Data Fig. 2a), resulting in the localization of TFB-rich regions within the binary composite film. Based on these observations, it can be inferred that the structure of the ternary nanocomposite originates from the large miscibility difference of QDs and TFB as compared to SEBS-g-MA. During the spin-coating of the ternary composite, as the solvent continues to evaporate, TFB tends to be separated from the composite film and form TFB-rich regions, similar to the case of the binary-blended films of TFB/SEBS-g-MA. In contrast, the homogeneous mixture of QDs and SEBS-g-MA is still suspended in the solution state. After the entire solvent evaporates, TFB-rich islands can be precipitated at the bottom part of the blended film. The generation of TFB-rich islands is further supported by the increase in the size of the bottom islands as more TFB is added, as shown in Extended Data Fig. 2b.

We analysed a depth profile of the stretchable EML with a composition of $Q_{100}S_{10}T_5$ by using X-ray photoelectron spectroscopy (XPS) (Fig. 1d). The atomic fractions of carbon and nitrogen sharply increase near the bottom of the film, indicating an increase of the TFB composition near the bottom. The atomic fractions for the elements of QDs such as Zn, S and Cd decrease near the bottom, indicating that the relative amount of CdSe/ZnS QDs decreases. The vertically separated internal structure of the stretchable EML is particularly effective for the stretchable QLEDs with a normal device structure—a bottom-to-top structure: bottom anode, hole injection layer (HIL), hole transport layer, QDs, electron transport layer and top cathode—as hole injection from the bottom interface can be facilitated by the TFB-rich islands.

Observation of the EML morphology under stretching

We carefully optimized the composition of the stretchable EML to achieve both high stretchability without crack generation and efficient charge transport to embedded QDs. As the weight fraction of the elastomer increases, for instance, the mechanical resilience of the composite

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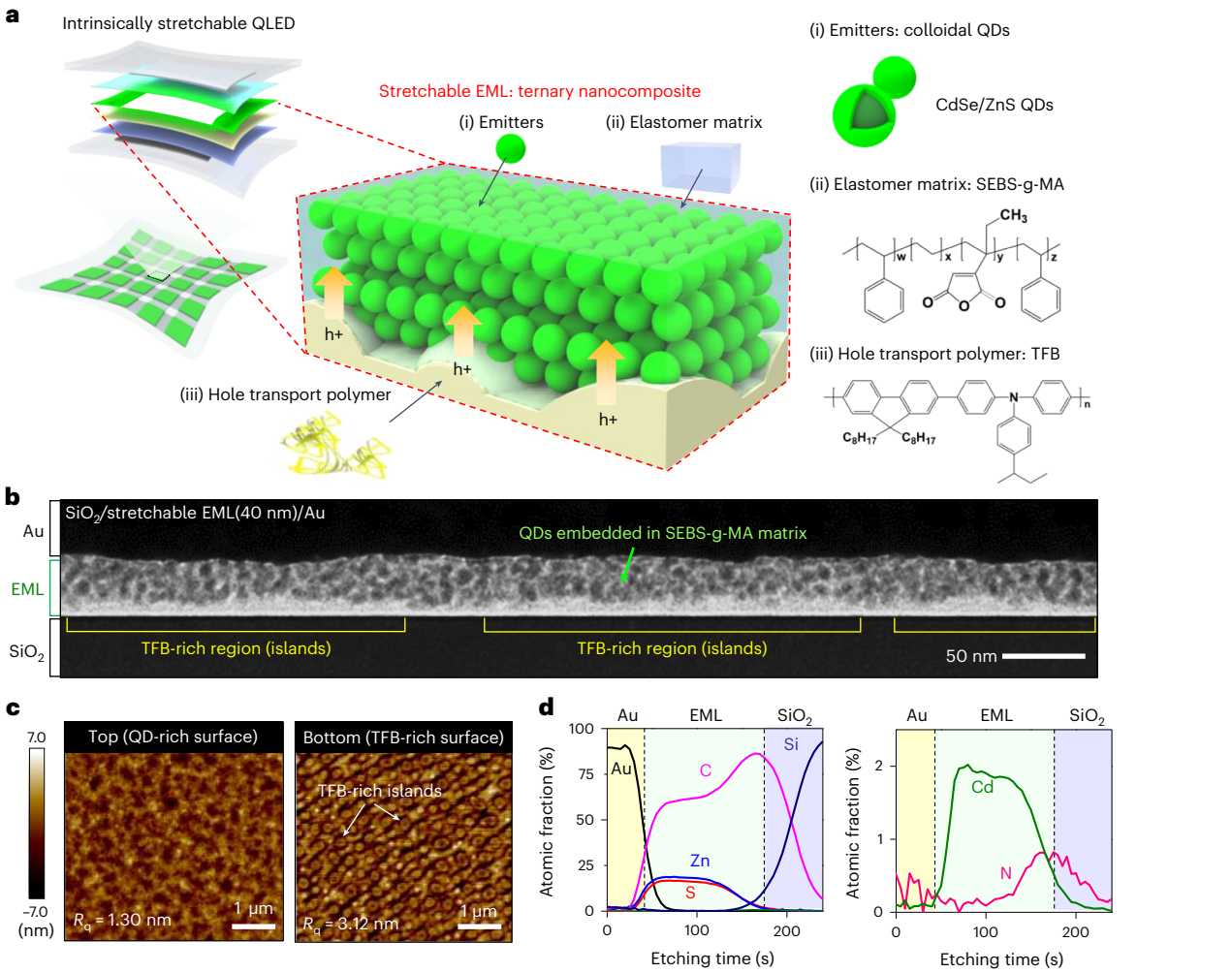

Fig. 1 | Characterization of stretchable EML that consists of QDs, an elastomer and a charge transport polymer. a, Schematic illustration of stretchable EML (centre) for intrinsically stretchable QLED (left). The stretchable EML is a ternary composite of QDs, SEBS-g-MA and TFB (right). **b**, Cross-sectional TEM image of stretchable EML ($Q_{100}S_{10}T_5$). **c**, AFM topography images of the top (left) and bottom (right) side of stretchable EML ($Q_{100}S_{10}T_5$). **d**, XPS depth profiling data of stretchable EML ($Q_{100}S_{10}T_5$). Compared to the atomic fraction of C, Zn and S (left), the fraction of Cd and N (right) is relatively low (less than 2%).

film is improved, but the charge injection to QDs is rather impeded by the elastomer. To understand the effect of the elastomer fraction on the mechanical properties of the stretchable EML films, their surface morphology under uniaxial strains was observed to monitor crack formation in the stretched films. The composite films were uniaxially stretched by following the 'stretch-and-transfer' process (Extended Data Fig. 3a). Three types of films with different compositions, including bare QD, $Q_{100}S_5T_5$ and $Q_{100}S_{10}T_5$ were prepared. After 50% uniaxial strain was applied, the morphology was observed with atomic force microscopy (AFM) (Fig. 2).

When the uniaxial strain was applied to the bare QD film, numerous microcracks were formed on its surface (Fig. 2a–c). Because there is no strong binding force between the nanoparticles, the cracks generated on the surface propagate across the entire film. In contrast, the composite film is more elastic, as the tensile stress can be released by the elastomer matrix. In the case of the $Q_{100}S_5T_5$, the 50% stretched film exhibits a partially cracked structure (Fig. 2d–f). When the weight fraction of the elastomer reached a critical concentration of 10 wt% (that is, $Q_{100}S_{10}T_5$), the top and bottom surfaces of the stretched film showed smooth surface morphologies without cracks, indicating that the composite film finally becomes fully stretchable (Fig. 2g–i).

The stretchable $Q_{100}S_{10}T_5$ film undergoes compression perpendicular to the stretching direction due to Poisson's effect, exhibiting thickness decrease under uniaxial strain (Fig. 2j). Interestingly, the compression did not occur evenly across all regions of the film, because the film has locally segregated TFB-rich regions at the bottom side. When subjected to stretching forces, in Region A, the majority of the compressive deformation occurs in the upper layer of the QD/SEBS-g-MA mixture, primarily because of the relatively high rigidity of the TFB-rich islands. Conversely, in Region B, which is mainly composed of the QD/SEBS-g-MA mixture, the entire film becomes fully compressed (Fig. 2g) Therefore, the uneven compression rate between Region A and Region B (Fig. 2d,g) resulted in an embossed structure with an amplitude of 5–8 nm, in accordance with the arrangement of TFB-rich islands at the bottom. The bottom surface of the $Q_{100}S_5T_5$ film showed partial cracks, whereas that of the $Q_{100}S_{10}T_5$ film showed no cracks but formed an embossed structure of the TFB-rich islands (Extended Data Fig. 3b,c).

During the uniaxial stretching of the composite film from 0% strain to 10, 30 and 50% strain, we analysed the interparticle distance inside the film by using grazing-incidence small-angle X-ray scattering (GISAXS) analysis (Fig. 2k and Extended Data Fig. 4a)[32,33]. According to the GISAXS data in the in-plane and out-of-plane direction (Extended Data Fig. 4b,c), both the horizontal and vertical inter-QD distances were not changed substantially when the film was stretched. The horizontal and vertical interplanar distances (d_{xy} and d_z) are 10.10 nm and 8.38 nm, respectively, at 0% strain, 10.31 nm and 8.05 nm at 10%, 10.15 nm and

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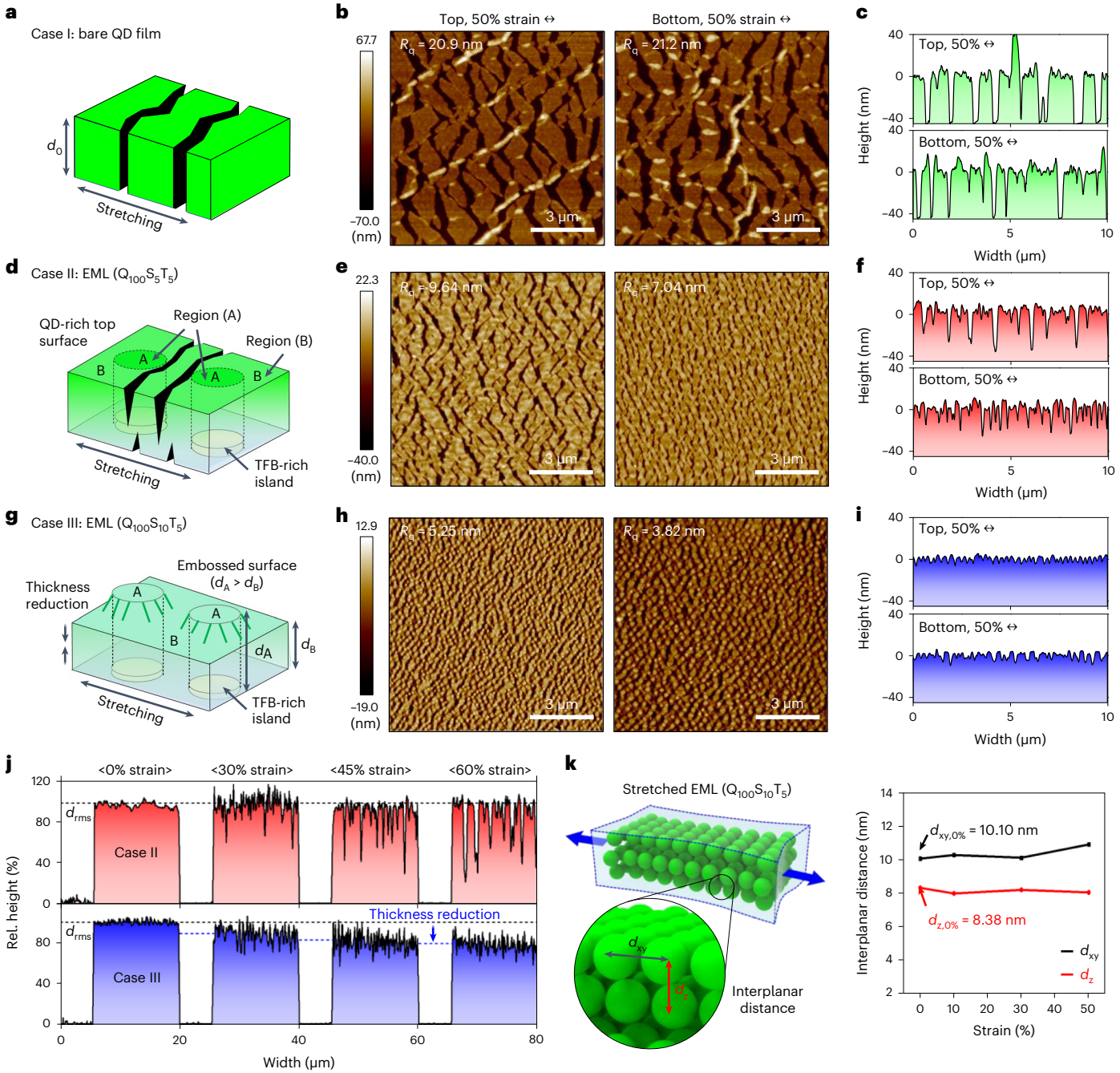

Fig. 2 | Mechanical properties of stretchable EML. a,**d**,**g**, Schematic illustrations of three types of films stretched by 50%: bare QD film (**a**), stretchable EML ($Q_{100}S_5T_5$) (**d**) and stretchable EML ($Q_{100}S_{10}T_5$) (**g**). **b**,**e**,**h**, AFM topography images of the top (left) and bottom (right) surfaces of the stretched films of **a** (**b**), **d** (**e**) and **g** (**h**). **c**,**f**,**i**, Height profiles of the top and bottom surfaces of the stretched films of **a** (**c**), **d** (**f**) and **g** (**i**). **j**, Height profiles of the stretched EML $Q_{100}S_5T_5$ (Case II, upper, red) and stretched EML $Q_{100}S_{10}T_5$ (Case III, lower, blue). Elastomeric compression occurs in stretched EML $Q_{100}S_{10}T_5$. **k**, Schematic illustration (left) and interplanar distance between embedded QDs (right) within the stretched EML ($Q_{100}S_{10}T_5$). Rel., relative.

8.26 nm at 30% and 10.93 nm and 8.11 nm at 50%. Thus, the average particle density of embedded QDs inside the stretched film remained essentially unchanged. This is attributed to the negligible change of the entire volume of the strained film, indicating the elastic deformation of the film (Extended Data Fig. 4d).

Enhanced hole injection properties of the stretchable EML

The engineered stretchable EML has advantages not only in soft mechanical properties but also in facile hole transport through the bottom-located TFB-rich regions (Fig. 3). To understand the behaviour of carriers, steady-state and time-resolved photoluminescence (PL) of the stretchable EML thin films with different TFB weight fractions—namely, 0, 3, 5 and 10 wt% of QDs—were measured (Fig. 3a,b). The 480 nm laser (2.6 eV) excitation used in PL analyses selectively excites green QDs, but not TFB. Even with a TFB weight fraction of up to 10 wt%, the emission spectra of the composite films exhibited no shift. The FWHM of the PL spectra of various stretchable EMLs was constant (~19 nm). The PL properties of the EML nanocomposite in the solution state are provided in Extended Data Fig. 5a,b.

The time-resolved PL data revealed that increasing the TFB content in the stretchable EML leads to a slight decrease in the carrier

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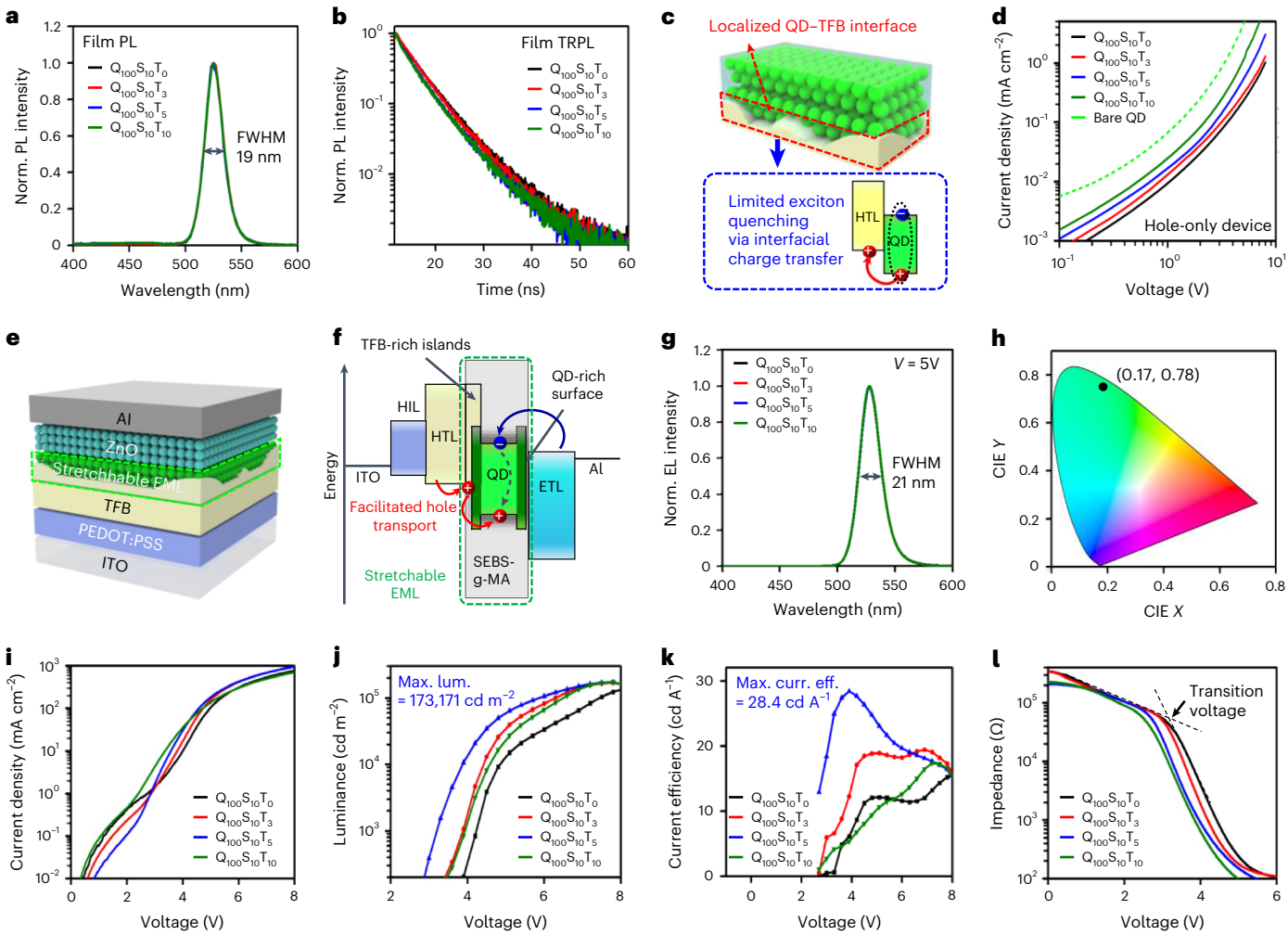

Fig. 3 | Optoelectronic properties of stretchable EML. a,b, Steady-state (**a**) and time-resolved (**b**) photoluminescence properties of stretchable EML films with various TFB weight fractions under 480 nm excitation. **c**, The localized QD–TFB interface results in reduced exciton quenching in the stretchable EML film, which is attributed to the interfacial charge transfer process. **d**, Current densities of hole-only devices. **e**–**l**, Characterization of the QLEDs using the stretchable EMLs with different TFB contents: device structure (**e**), energy band diagram (**f**), EL spectra (**g**), Commission Internationale de l'Éclairage coordinates (CIE) (**h**), *J*–*V* curves (**i**), *L*–*V* curves (**j**), current efficiency (**k**) and impedance–voltage data (**l**). Norm., normalized; TRPL, time-resolved PL; Max. curr. eff., maximal current efficiency.

lifetime. The decreased carrier lifetime is attributed to quenching of generated excitons, which is primarily caused by the interfacial charge transfer between QDs and TFB[34,35]. However, because of the unique internal structure of the stretchable EML, namely, the localization of the TFB-rich islands at the lower part of the film, the QD–TFB interface, which is a major exciton quenching site, was not substantially expanded even with higher TFB contents, resulting in a small reduction in the carrier lifetime (Fig. 3c). Note that the PL decay of the stretchable EML ($Q_{100}S_{10}T_5$) is slower than that of the QD–TFB binary blend ($Q_{100}T_5$) with the same weight ratio of QD to TFB (Extended Data Fig. 5c). The QD–TFB binary blend film has distributed interfaces between QD and TFB, resulting in a faster decay of the carrier lifetime via the interfacial exciton quenching process. (Extended Data Fig. 5d–f).

Next, we investigated the hole injection effect of the stretchable EML by comparing current densities of hole-only devices, namely, indium tin oxide (ITO), poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS, 30 nm), TFB (20 nm), stretchable EML (35 nm), MoO_3 (5 nm) and Au (100 nm) (ref. 36). As TFB was added to the composites, the hole current densities showed an increasing trend, indicating that the TFB-rich islands located at the bottom of the stretchable EML function as an efficient hole transport pathway (Fig. 3d).

Then, we compared the electroluminescence properties of QLEDs made of stretchable EML films with different TFB weight fractions. These QLEDs were fabricated on the rigid ITO substrates with the normal structure, that is, ITO, PEDOT:PSS (30 nm), TFB (20 nm), stretchable EML (35 nm), ZnO (30 nm) and Al (60 nm) (Fig. 3e,f)[37]. The unique internal structure of the stretchable EML, where the TFB-rich islands serve as the hole transport pathways, contributes to the charge balance. These QLEDs exhibited outstanding colour reproducibility, such as the narrow FWHM of 21 nm (Fig. 3g) and the deep green colour on the colour coordinate system of the Commission Internationale de l'Éclairage (Fig. 3h). The emission wavelength of the QLED remained stable even when the TFB content was increased to 10 wt%, supporting that the radiative decay of excitons mainly occurs at the QDs. The excellent colour reproducibility is another advantage of the QD-based stretchable EML, compared to the previously reported stretchable organic EMLs.

Compared to the binary composite of QD and SEBS-g-MA ($Q_{100}S_{10}T_0$), the addition of TFB to the composites resulted in the increase of current density, luminance and current efficiency of QLEDs (Fig. 3i–k). The optimal content of TFB was 5 wt%, showing the maximum brightness of 173,171 cd m^{-2} and the maximum current efficiency of 28.4 cd A^{-1}. Such excellent EL performance was attributed

to the balanced injection of electrons and holes into the QDs, which is facilitated by the enhanced hole injection through the bottom-located TFB-rich islands. The impedance spectroscopy was also carried out to analyse the facilitated hole transport properties of the stretchable EML (Fig. 3l)[38–40]. The transition voltages, at which the impedance began to dramatically reduce, gradually decreased as the TFB contents increased from 0 to 10 wt%. The impedance analysis data strongly support that the TFB content in the stretchable EML fosters the hole injection to QDs, lowering the driving voltage of the device.

When the TFB content reached 10 wt%, however, the performance of the QLED began to deteriorate, even though there was still a trend of decreasing turn-on voltage. The decline in luminous performance with the excessive TFB content is attributed to the occurrence of the leakage current bypassing QDs, which does not contribute to the radiative decay of excitons. The overflows of bypassing currents may be caused by the uneven surface morphologies of $Q_{100}S_{10}T_{10}$. As shown in Extended Data Fig. 2b, $Q_{100}S_{10}T_{10}$ has a non-uniform and rough surface characterized by substantial macroscopic phase separation. The embossed part of $Q_{100}S_{10}T_{10}$ may induce the leakage current bypassing QDs, resulting in a decreased device efficiency.

Intrinsically stretchable QLEDs

Based on the approach, the optimized stretchable EML ($Q_{100}S_{10}T_5$) was readily applied to the fabrication of intrinsically stretchable QLEDs. Figure 4a shows an exploded view of the intrinsically stretchable QLEDs with material information for each layer. All components of the QLED were carefully optimized, considering their mechanical as well as electrical properties. For the stretchable bottom electrodes, we used a patterned silver nanowire (Ag NW) mesh embedded in ultraviolet (UV)-curable polyurethane acrylate (PUA) resin (Fig. 4a(i)). Then, a stretchable charge spreading layer, consisting of highly conductive PEDOT:PSS modified by dimethyl sulfoxide (DMSO, 5 wt%) and surfactant (FS-31, 1 wt%), was deposited onto the bottom electrodes (Fig. 4a(ii))[41,42]. By adding DMSO and the surfactant, the electrostatic interaction between PEDOT and PSS became disrupted and thus the phase separation occurred, where a linear PEDOT-rich domain contributes to mechanical stretchability[43,44]. Next, semiconducting PEDOT:PSS, modified with 1 wt% FS-31, was spin-coated onto the conductive PEDOT:PSS layer as a stretchable HIL (Fig. 4a(iii)). The modified PEDOT:PSS can be stretched up to 60%, without severe crack generation (Extended Data Fig. 6b).

For the hole transport and emission layers, a bilayer film of TFB and $Q_{100}S_{10}T_5$ was spin-coated on the stretchable HIL (Fig. 4a(iv)). Although TFB has a low crack-onset strain of less than 25% (ref. 6), the double-layered structure (TFB (20 nm) and stretchable EML (35 nm), coated on elastomer substrate) does not show cracks up to 60% of uniaxial strain (Extended Data Fig. 6c,d). For the electron transport layer, the inorganic-organic hybrid of ZnO nanoparticles and poly(9,9-bis(3′-(N,N-dimethyl)-N-ethylammoinium-propyl-2,7-fluorene)-alt-2,7-(9,9-dioctylfluorene))dibromide (PFN-Br) was used (Fig. 4a(v)). While the pristine ZnO film exhibits severe crack generation across the entire film surface during its stretching, the ZnO/PFN-Br composite film has a microcracked structure under the same condition (Extended Data Fig. 6e). The double-layered structure (ZnO/PFN-Br (40 nm) and stretchable EML (35 nm), coated on elastomer substrate) exhibits smooth surface morphology without crack generation up to 60% of strain, indicating that the ZnO/PFN-Br film does not influence the stretchability of stretchable EML (Extended Data Fig. 6f). The top electrode consists of a spray-coated Ag NW mesh, a thin Ag film and a spray-coated liquid metal (LM) layer (Fig. 4a(vi)). Here, the LM layer improves not only charge injection by covering the void area of the Ag NW mesh but also stretchability by maintaining the percolation network even under 100% strain[45]. A thin Ag film was used for facile spray-coating of the LM layer, as the coverage of the spray-coated LM layer was dramatically increased on the Ag film. Finally, the device was encapsulated with UV-curable PUA resin (Fig. 4a(vii)). The detailed fabrication process is described in Extended Data Fig. 6a and Methods. Also, the resistance change data of stretchable anodes and cathodes under the applied strain are provided in Extended Data Fig. 6g.

Figure 4b shows luminous performance of the intrinsically stretchable QLEDs. The device showed the maximum luminance of 15,170 cd m^{-2} and turn-on voltage of 3.2 V. This is, to our knowledge, the record-high brightness at such a low driving voltage of 6.2 V, which is superior to the previously reported d.c.-driven, intrinsically stretchable organic LECs and organic LEDs (Fig. 4c and Extended Data Table 1). The maximum EQE of the stretchable device is 1.2%, while the maximum current efficiency reaches 5.0 cd A^{-1} at the driving voltage of 5.3 V (Extended Data Fig. 6h). Furthermore, there is no degradation of the brightness of the intrinsically stretchable QLEDs regardless of the applied strain up to ~50% (Fig. 4d,e). The initial increase of the brightness and current density (<30% strain) is primarily due to the thickness reduction of the device, which results in the increase of net voltage across the stretchable EML. The stability of the devices, including their lifetime and effect of the cyclic stretching test, was also studied (Extended Data Fig. 6g,h). The intrinsically stretchable QLED has an operational lifetime (the time taken for the brightness of the LED to decrease to 50% of the initial value) of ~20 min for the initial luminance of 100 cd m^{-2}, and the device can endure ~200 cycles of 30% stretching, without major degradation of luminous performance. Further details of the stretching demonstrations are shown in Extended Data Fig. 7a–d.

Based on the device design of the stretchable QLED, the intrinsically stretchable 5 × 5 PM-QLED arrays were fabricated. The PM-QLED arrays used a common cathode configuration for their operation, of which the external circuits sequentially controlled one row (cathode) and all columns (anode) for each cycle[46]. A customized circuit based on Arduino Teensy 3.6 was used for the PM operation of the QLED arrays (Extended Data Fig. 8). The fabricated QLED array had high mechanical deformability, exhibiting stable device operation even under various stretching deformations, such as deformation from a flat mode to a convex mode (Fig. 4f) or stretching by a uniaxial strain of 30% (Fig. 4g). The intrinsically stretchable PM-QLED array was also capable of displaying specific patterns, such as a line pattern (Fig. 4h), a letter 'A' or a character '♥' (Fig. 4i). Further details regarding the operation of the intrinsically stretchable PM-QLED arrays are shown in Supplementary Video 1 and Extended Data Fig. 7e.

The stretchable EML can be readily patterned by using transfer printing process[43], showing its potential for the fabrication of full-colour displays. Using a similar strategy, red- and blue-emitting stretchable EMLs were also formed by replacing green QDs with red and blue QDs. According to the *J*–*V*–*L* characteristics of QLEDs (ITO, PEDOT:PSS, TFB, stretchable EML, ZnO and Al) using red- and blue-emitting EMLs, the addition of TFB to the composite has brought the increase of device performance for both red and blue QDs, exhibiting the versatility of the strategy (Extended Data Fig. 9a,b). The thickness of the red- and blue-emitting stretchable EMLs was configured as 35 nm. The maximum current efficiency of the red-emitting intrinsically stretchable QLED is increased from 0.8 cd A^{-1} ($Q_{100}S_{10}$) to 1.8 cd A^{-1} ($Q_{100}S_{10}T_5$), while that of the blue-emitting QLED is increased from near zero ($Q_{100}S_{10}$) to 0.5 cd A^{-1} ($Q_{100}S_{10}T_{10}$).

Full-colour images, which consist of RGB pixels of stretchable EMLs, were sequentially printed on a stretchable substrate via aligned multiple transfer printing of the stretchable EMLs (Fig. 5a and Extended Data Fig. 9c–h). The resulting PL images with RGB stretchable EML patterns are shown in Fig. 5b and Extended Data Fig. 9i. The PL image can withstand 50% of uniaxial strain without any cracks (Fig. 5c and Extended Data Fig. 9j). Moreover, under the biaxial strain of 50%, the image can be enlarged without any distortion (Fig. 5d and Extended Data Fig. 9k). The facile patterning of the stretchable EML enables the fabrication of multicolour intrinsically stretchable QLEDs (Fig. 5e,f and Supplementary Video 2). The red and green stretchable EMLs were

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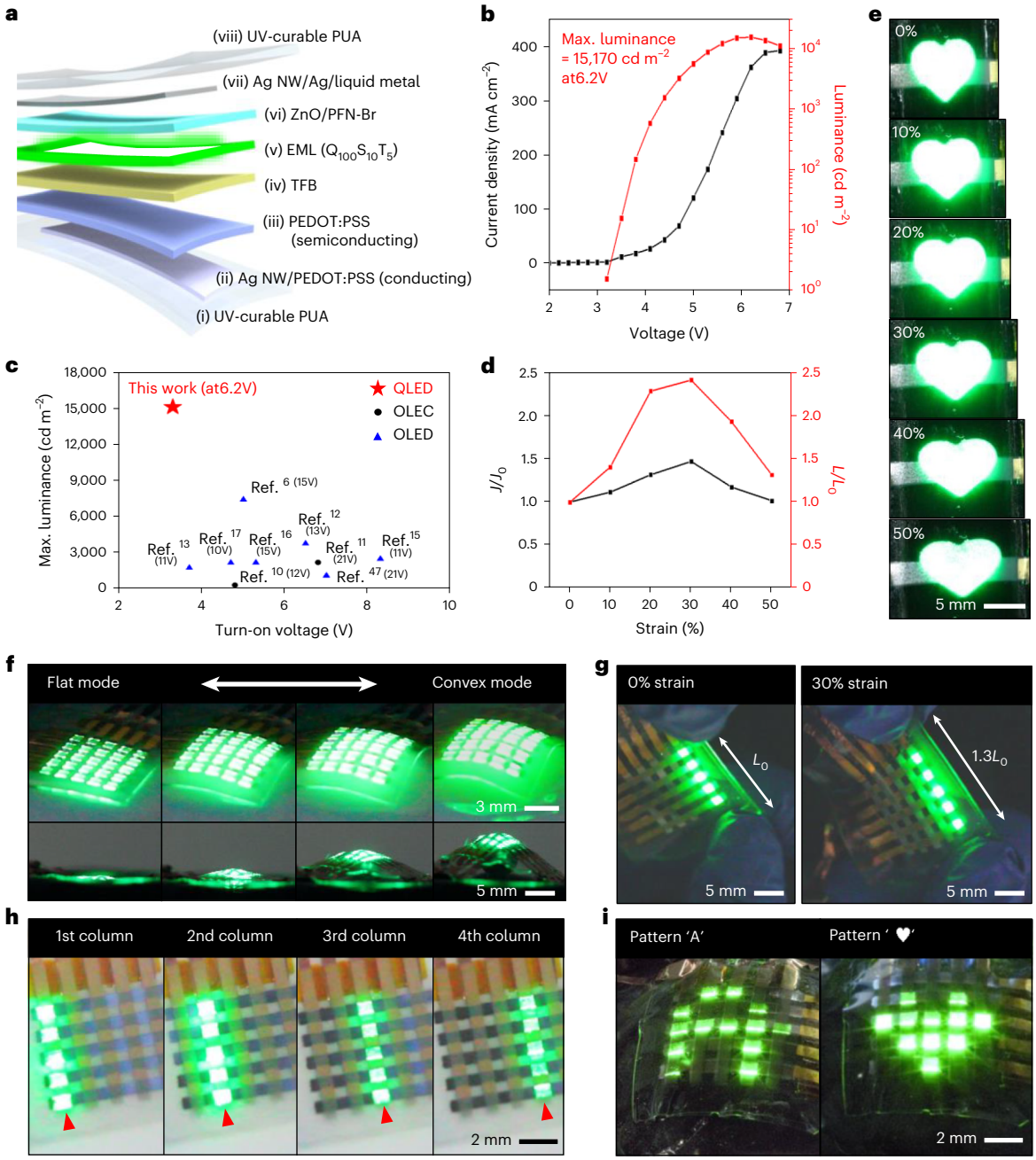

Fig. 4 | Intrinsically stretchable QLEDs. a, Device structure of the intrinsically stretchable QLED. **b**, *J*–*V*–*L* data of the intrinsically stretchable QLED. **c**, Comparison of previously reported d.c.-driven intrinsically stretchable organic LECs and organic LEDs with our work, in regard to maximum luminance and turn-on voltage. The voltage values in parentheses represent the voltage required for driving maximum brightness in each reference. **d**, Strain-dependent current density and luminance change under strains up to 50%. **e**, Photographs of intrinsically stretchable QLED stretched up to 50%. **f**, Photographs of intrinsically stretchable PM-QLED array, which shows a series of deformations from the flat mode to the convex mode. **g**, Photographs of PM-QLED array, stretched by 30% strain. **h**, Photographs of PM-QLED array, driven in column by column. **i**, Photographs of PM-QLED array, displaying 'A' (left) and '♥' (right). J_0, current density at 0% strain; L_0, luminance at 0% strain; Max., maximum.

transfer-printed on the hole transport layer to fabricate multicolour emitting devices, exhibiting stable performance under mechanical stretching of ~50% strain. The successful fabrication of the multicolour intrinsically stretchable PM-QLED array demonstrates the feasibility of the stretchable full-colour displays (Fig. 5g).

Conclusions

We have reported the development of an intrinsically stretchable EML that is based on a ternary composite consisting of QDs, an elastomer (SEBS-g-MA) and a hole transport polymer (TFB). By optimizing the ternary composite composition, the EML can be stretched up to 50% strain without forming cracks and showing minimal change of the interparticle distance between embedded QDs. In addition, TFB-rich regions at the bottom of the composite facilitate hole injection to the QDs, promoting exciton recombination in the embedded QDs. The intrinsically stretchable EML was used to create green-emitting intrinsically stretchable QLEDs that exhibit high brightness (15,170 cd m^{-2} at 6.2 V) and mechanical stretchability (~50%). We also fabricated a passive-matrix array of intrinsically stretchable QLEDs, which exhibits stable operation under various mechanical deformations. Furthermore,

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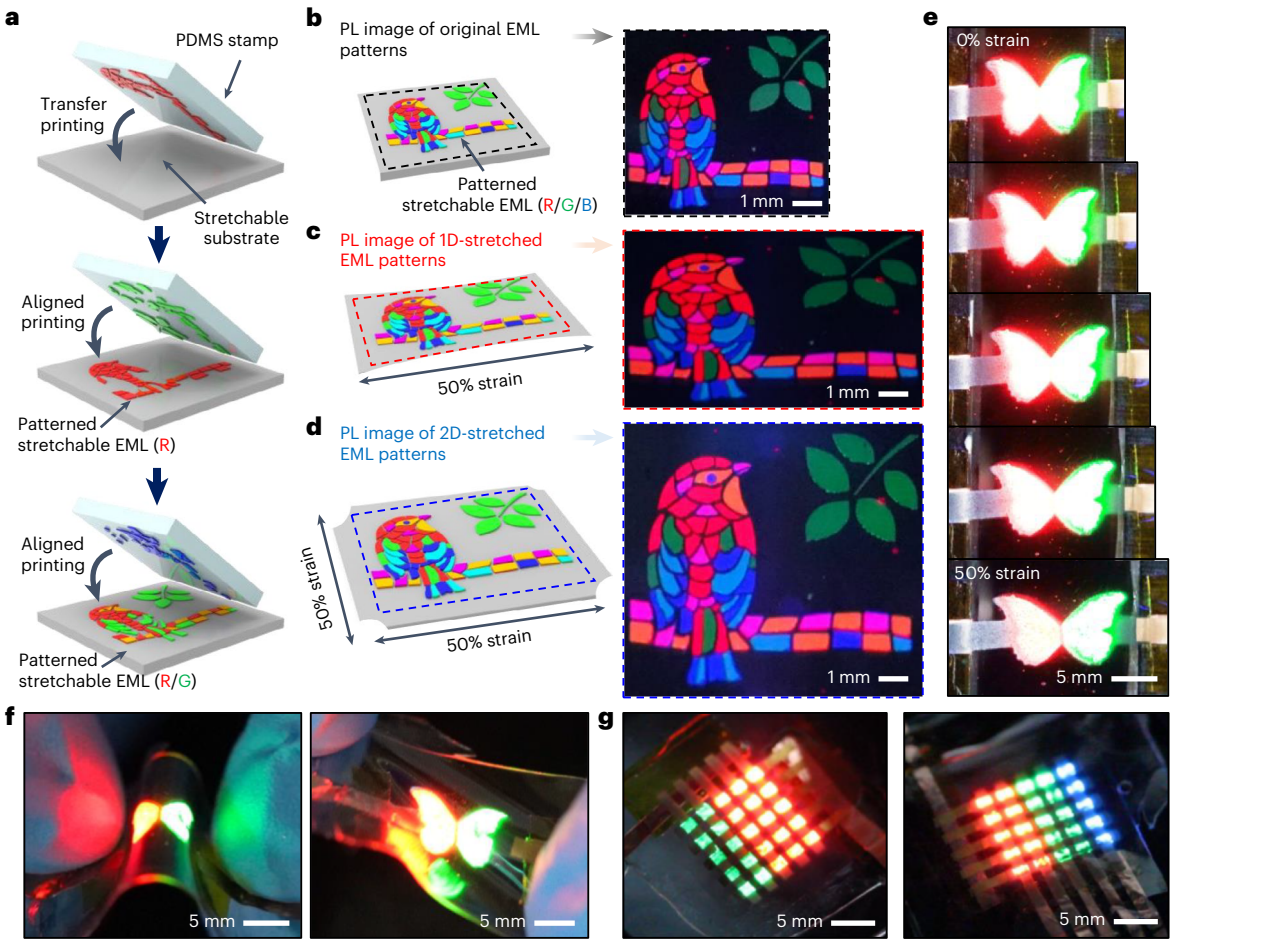

Fig. 5 | Patterning of stretchable EML for multicolour emitting stretchable displays. a, Patterning procedure of the RGB stretchable EML patterns via the transfer printing technique. **b**–**d**, Schematic (left) and PL images (right) of printed RGB stretchable EML patterns. The printed patterns are shown in their original form (**b**), under 50% uniaxial strain (**c**) and under 50% biaxial strain (**d**). **e**, Photographs of the multicolour intrinsically stretchable QLED under 50% of uniaxial strain. **f**, Various deformation modes of the multicolour intrinsically stretchable QLED. **g**, Photographs of multicolour intrinsically stretchable PM-QLED arrays. 1D, one-dimensional; 2D, two-dimensional.

by sequentially printing red, green and blue QDs via aligned intaglio transfer printing, intrinsically stretchable pixels emitting RGB colours can be readily patterned on a stretchable substrate. This enabled the fabrication of full-colour passive-matrix QLED arrays.

The luminous performance, lifetime and stretchability of the intrinsically stretchable QLEDs could be further enhanced through the development of functional materials for the internal components of the device. For instance, elastomeric polymers with excellent electrical properties can overcome the trade-off between the stretchability and charge injection characteristics[17]. Furthermore, by engineering the interaction between adjacent polymers and QD surfaces/ligands, charge transport to the QDs could be improved. Finally, the development of intrinsically stretchable charge transport layers and electrodes with minimal resistance increase under applied strain will be essential for achieving high-performance intrinsically stretchable QLEDs.

Methods

Preparation of the stretchable QD-polymer composite

Three types of QDs (red, green and blue) were synthesized by using a method modified from a previous report[47]. Cadmium oxide (CdO, >99.99%), selenium powder (Se, 99.999%), zinc acetate ($Zn(OAc)_2$, 99.99%), sulfur powder (S, 99.99%), oleic acid (OA, 90%), 1-octadecene (1-ODE, 90%), trioctylphosphine (TOP, 90%), tributylphosphine (TBP, 95%), 1-octanethiol (98.5%), cyclohexane (anhydrous, 99.5%) and ethanol (anhydrous, 99.5%) were purchased from Sigma-Aldrich, USA. OA and 1-ODE were degassed at 120 °C for 2 h before their use. The as-synthesized QDs were treated with OA for the ligand exchange. Then, the QD solution (dispersed in toluene) was mixed with the SEBS-g-MA (purchased from Kraton, dispersed in toluene) and TFB solution (purchased from Lumtec, dispersed in toluene) by using a desired weight ratio of three materials. SBS (purchased from Kumho Petrochemical) and SEBS (purchased from Asahi Kasei) were both dispersed in toluene before their blending. The composite solution was sonicated for 2 h and filtered through a 200 nm pore polytetrafluoroethylene (PTFE) filter. A thin film of the ternary composite was then coated on the substrate via spin-coating. The thickness of the film was manipulated by controlling the speed of spin-coating.

Fabrication of intrinsically stretchable QLEDs

To fabricate the bottom electrode, an Si wafer treated with a self-assembly monolayer (Trichloro(octadecyl)silane, Sigma-Aldrich) was prepared and the Ag NW solution (0.1 wt% in isopropyl alcohol) was spray-coated onto the Si wafer in a patterned manner using shadow masks. The PUA resin was poured onto the Ag-NW-coated substrate and cured by UV light[48]. The Ag-NW-embedded PUA film was detached from the Si wafer, and the PEDOT:PSS (PH1000 modified with 5 wt% DMSO and 1 wt% of FS-31) solution was spray-coated in ambient conditions onto the Ag-NW-embedded surface of the PUA substrate. The modified

HIL (AI4083, Heraeus) with 1 wt% of FS-31, 30 nm), TFB (20 nm) and stretchable EML (35 nm) were successively deposited onto the stretchable anodes via spin-coating under Ar-filled glovebox condition. For the multicolour intrinsically stretchable QLEDs, stretchable EMLs with different colours were printed onto the device via multiple aligned transfer processes. After spin-coating the mixture of ZnO nanoparticles and PFN-Br solution (2.5 mg ml^{-1} in ethanol) to form 40 nm of electron transport layer film, the Ag NW solution was spray-coated in ambient conditions through shadow masks. Then, Ag thin film and LM were coated sequentially on top. LM was spray-coated in ambient conditions using shadow masks. Finally, the intrinsically stretchable QLED was encapsulated by the UV-curable PUA resin.

Patterning of the stretchable EML

The stretchable EML was patterned through the intaglio transfer printing process, previously reported in ref. 43. The stretchable EML was spin-coated on the 1-octadecyltrimethosysilane-treated silicon substrate (donor substrate), and a flat polydimethylsiloxane stamp was utilized to quickly pick up the stretchable EML. The picked-up layer was then conformally contacted on the intaglio trench with low pressure and slowly detached. The non-contacted stretchable EML layer remaining on the polydimethylsiloxane stamp can be transferred to the stretchable substrate or devices. The intaglio trenches were negatively carved on the Si substrate using a deep Si etcher (Versaline, PLASMA THERM).

Material and device characterization

Depth-XPS profiling was performed using an ESCALAB 250Xi installed at the Korea Basic Science Institute. The surface morphology of the composite film was analysed using atomic force microscopy (AFM, Bruker Dimension Icon). The samples of stretched films were prepared by 'stretch-and-transfer' process (Extended Data Fig. 3). TEM analysis was performed using the JEM-ARM200F microscope (JEOL) installed at the National Center for Inter-university Research Facilities at the Seoul National University and Titan Themis Z (Thermo Fisher Scientific). GISAXS analysis was performed at Pohang Accelerator Laboratory (PAL) using the 3C and 6D UNIST-PAL beam lines of Pohang Light Source-II. An X-ray beam (11.56 keV) is irradiated onto the $Q_{100}S_{10}T_5$ film at an angle of 0.15°. The current–voltage curves of intrinsically stretchable QLEDs were measured with a Keithley 2636 source metre, and the brightness was measured with a CS-2000 spectrophotometer in ambient conditions. For the case of hole-only devices and ITO-QLEDs, the devices were properly encapsulated with UV-curable resins.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding authors upon reasonable request.

Code availability

The source codes for Arduino 1.8.3 are available from the corresponding authors upon reasonable request.

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Author contributions

D.C.K., H.S., J. Yoo, J.K., J. Yang, M.K.C., T.H. and D.-H.K. designed the experiments, analysed the data and wrote the paper. J.K., H.H.S. and K.L. synthesized the quantum dots, made composite solutions and conducted optical analysis. D.C.K., H.S., J. Yoo, J.S.K., Y.K., C.C., D.J., C.P. and H.H. fabricated the intrinsically stretchable QLED and characterized the device performance. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Jiwoong Yang, Taeghwan Hyeon, Moon Kee Choi or Dae-Hyeong Kim.

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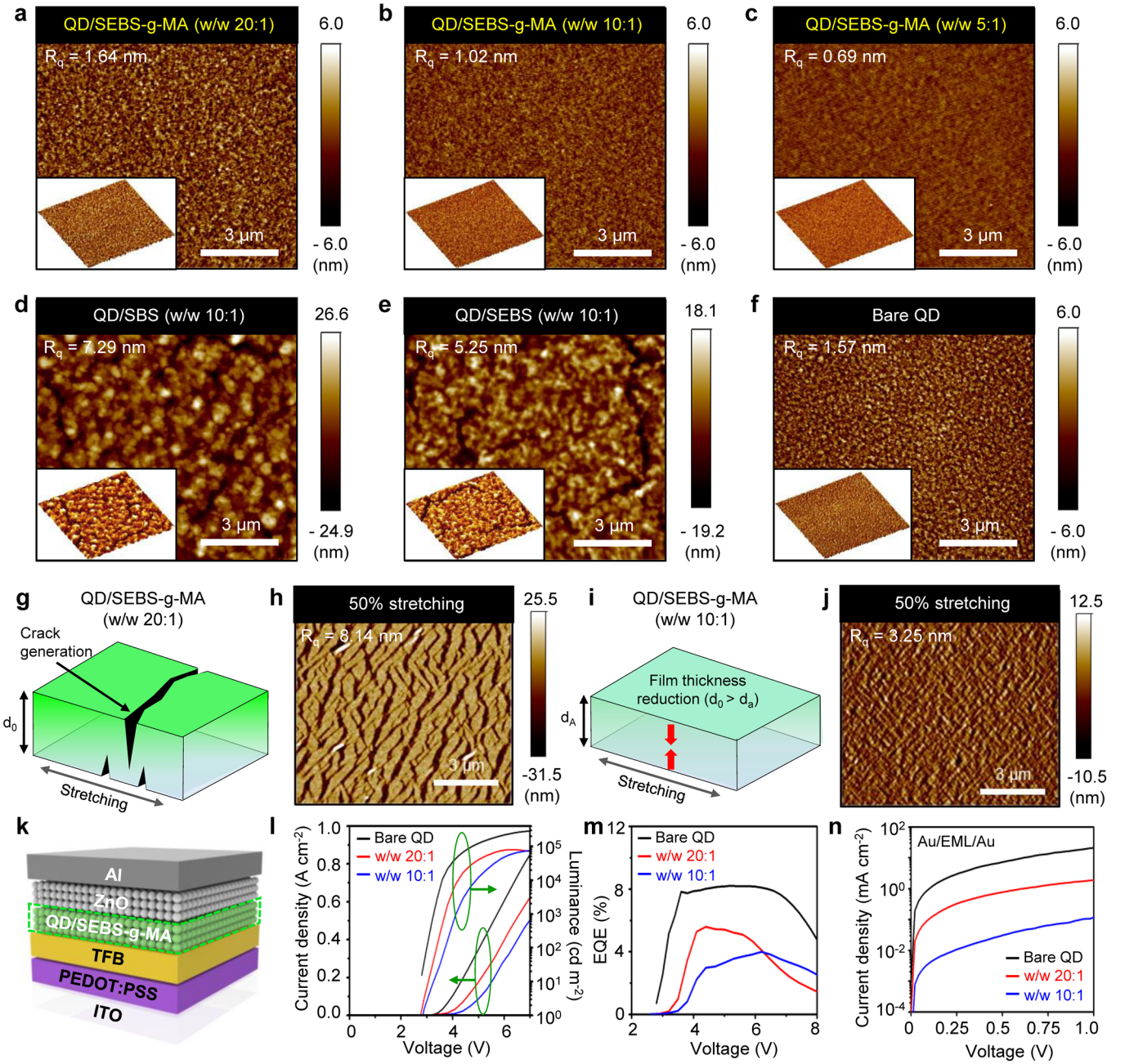

Extended Data Fig. 1 | Characterization of binary QD-elastomer composite films. a-f, AFM topography images and three-dimensional images (inset) of QD/SEBS-g-MA (w/w = 20:1) (a), QD/SEBS-g-MA (w/w = 10:1) (b), QD/SEBS-g-MA (w/w = 5:1) (c), QD/SBS (w/w = 10:1) (d), QD/SEBS (w/w = 10:1) (e), and bare QD (f) films. **g**,**h**, QD/SEBS-g-MA (w/w = 20:1) film under 50% of stretching. Schematic image (g) and the AFM image of the stretched film (h). **i**,**j**, QD/SEBS-g-MA (w/w = 10:1) film under 50% of stretching. Schematic image (i) and the AFM image of the stretched film. (j). **k-m**, Device structure of conventional QLEDs (k), *J*–*V*–*L* characteristics (l), and EQE data (m). In (l) and (m), three cases were compared, that is, bare QDs (black) or binary QD/SEBS-g-MA composites with different weight fractions (red for w/w = 20:1, or blue for w/w = 10:1). **n**, *J*–*V* characteristics of Au/EML/Au device were compared with bare QDs (black) or binary QD/SEBS-g-MA composites (red for w/w = 20:1, or blue for w/w = 10:1).

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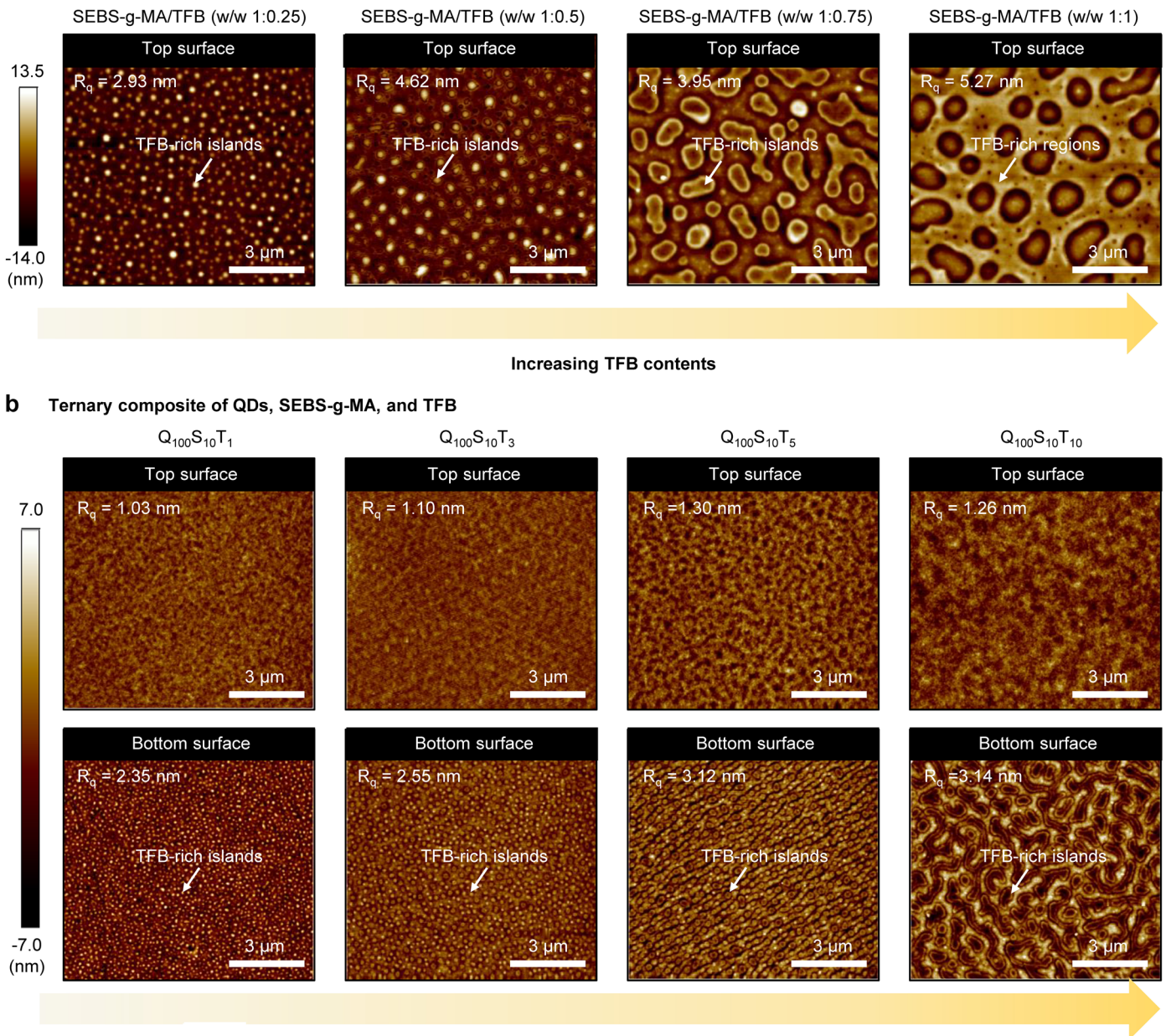

Extended Data Fig. 2 | Effect of the TFB weight fraction on the size of island-shaped domains. a, AFM topography images of binary composite films of SEBS-g-MA and TFB, with increasing TFB weight fractions. The size of island-shape domains increases as more TFB is added. **b**, AFM topography images of top and bottom surfaces of stretchable EML films with increasing TFB weight fractions. Similar to **a**, the size of island-shape domains increases as more TFB is added.

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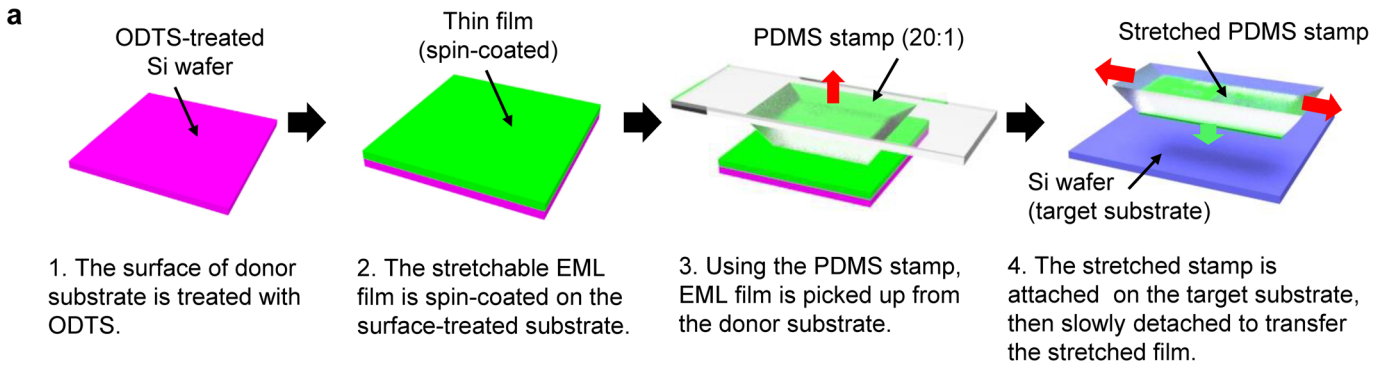

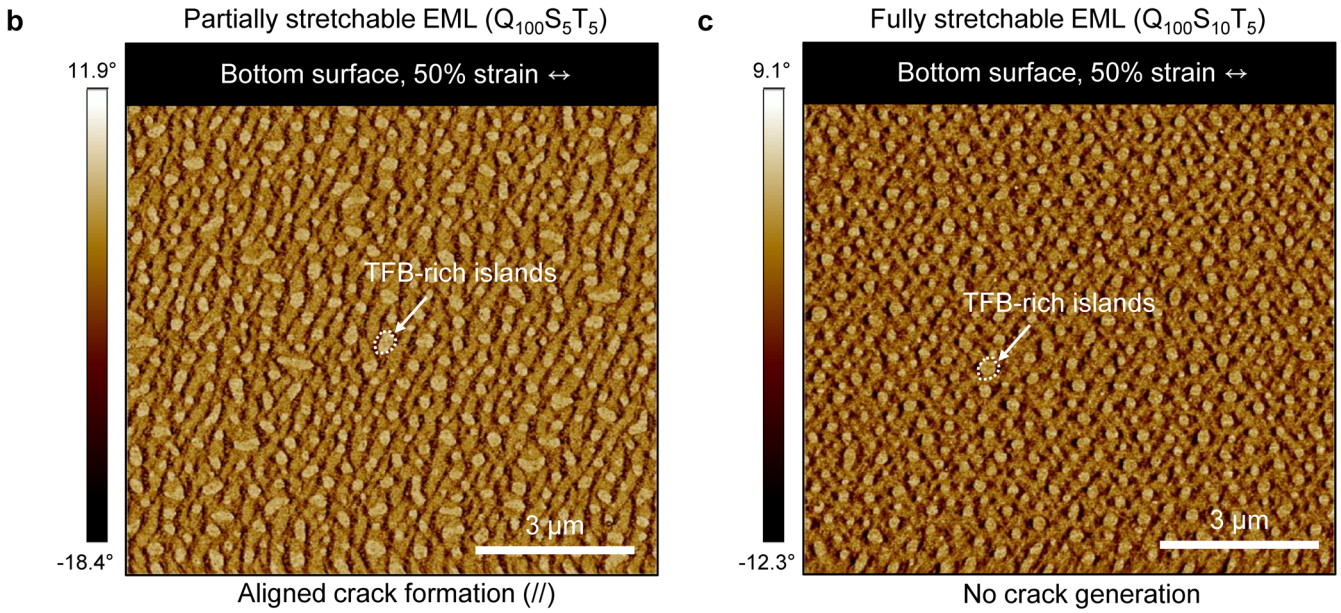

Extended Data Fig. 3 | Uniaxially stretched stretchable EML films. a, "Stretch-and-transfer" process to prepare samples of stretched films for the AFM analysis. **b**,**c**, AFM phase analysis of the stretched stretchable EML ($Q_{100}S_5T_5$) film (b) and stretchable EML ($Q_{100}S_{10}T_5$) film (c). Partial cracks are found around the TFB-rich islands in the stretchable EML ($Q_{100}S_5T_5$) film, while the embossed structure is formed in the stretchable EML ($Q_{100}S_{10}T_5$) film.

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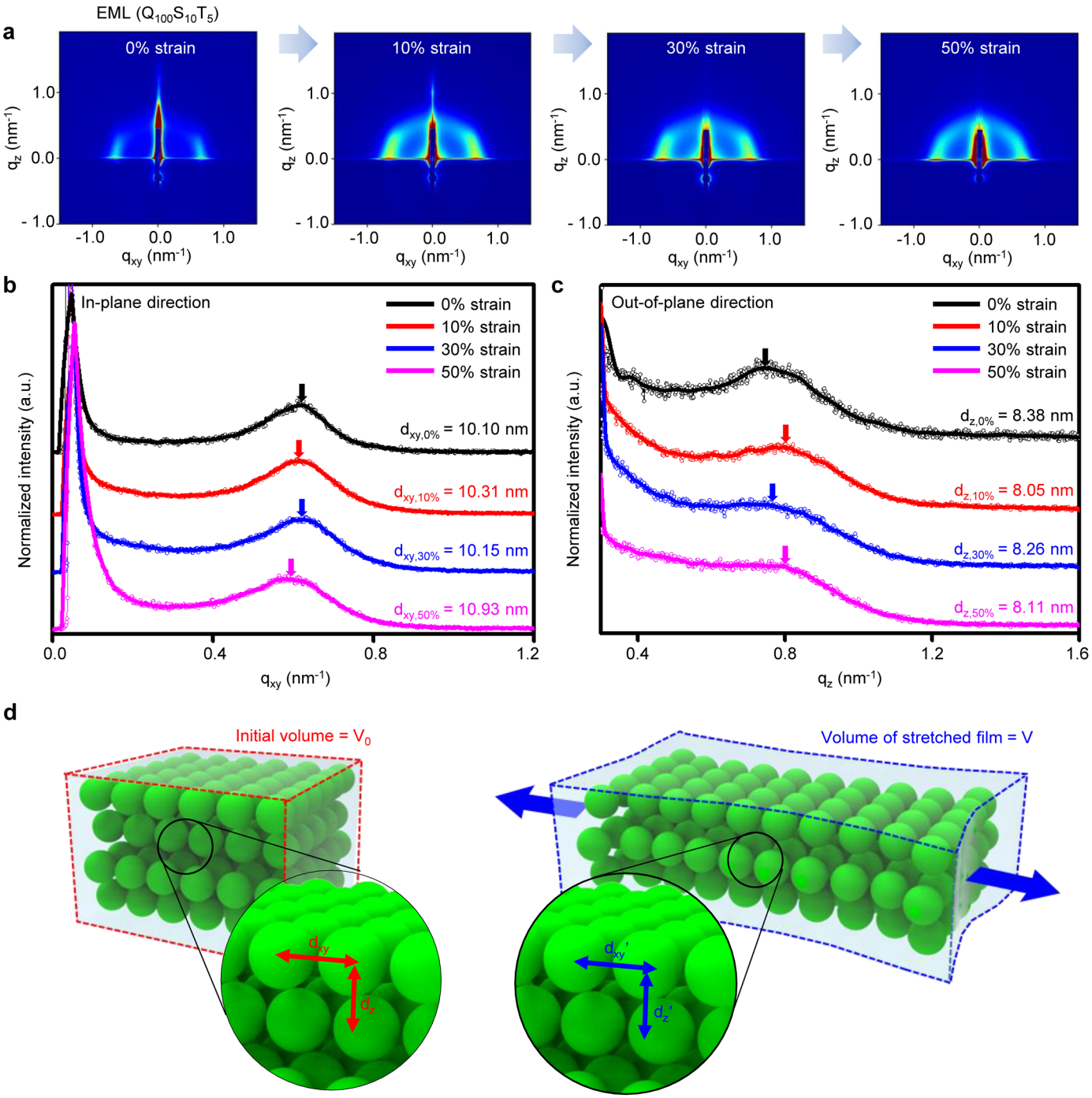

Extended Data Fig. 4 | The inter-particle distance of the stretchable EML ($Q_{100}S_{10}T_5$) film. a, Raw data of GISAXS measured with the films under strains of 0, 10, 30, and 50%. **b**,**c**, GISAXS analysis in the in-plane (b) and out-of-plane direction (c). **d**, Schematic illustrations of the stretchable EML film with the same volume and interplanar distances (*for example*, d_{xy}, d_z) before (left) and after (right) the stretching.

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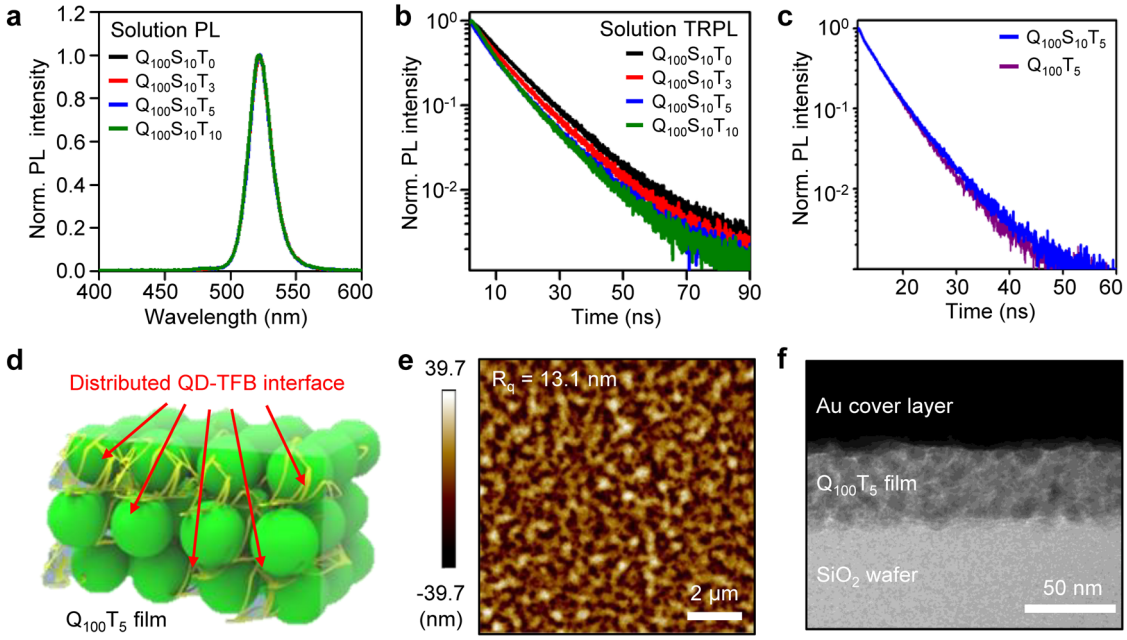

Extended Data Fig. 5 | Photoluminescence properties and exciton quenching of the stretchable EML. a,**b**, Photoluminescence (PL) properties of the stretchable EML solution with different TFB weight fractions under 480 nm excitation. Steady-state (a) and time-resolved PL (b). **c**, Time-resolved PL of the $Q_{100}S_{10}T_5$ film and $Q_{100}T_5$ film. **d-f**, Morphology of the QD/TFB binary blend film. Schematic illustration (d), AFM image (e), and cross-sectional TEM image (f) of the $Q_{100}T_5$ film, exhibiting the distributed QD-TFB interface inside the blended film.

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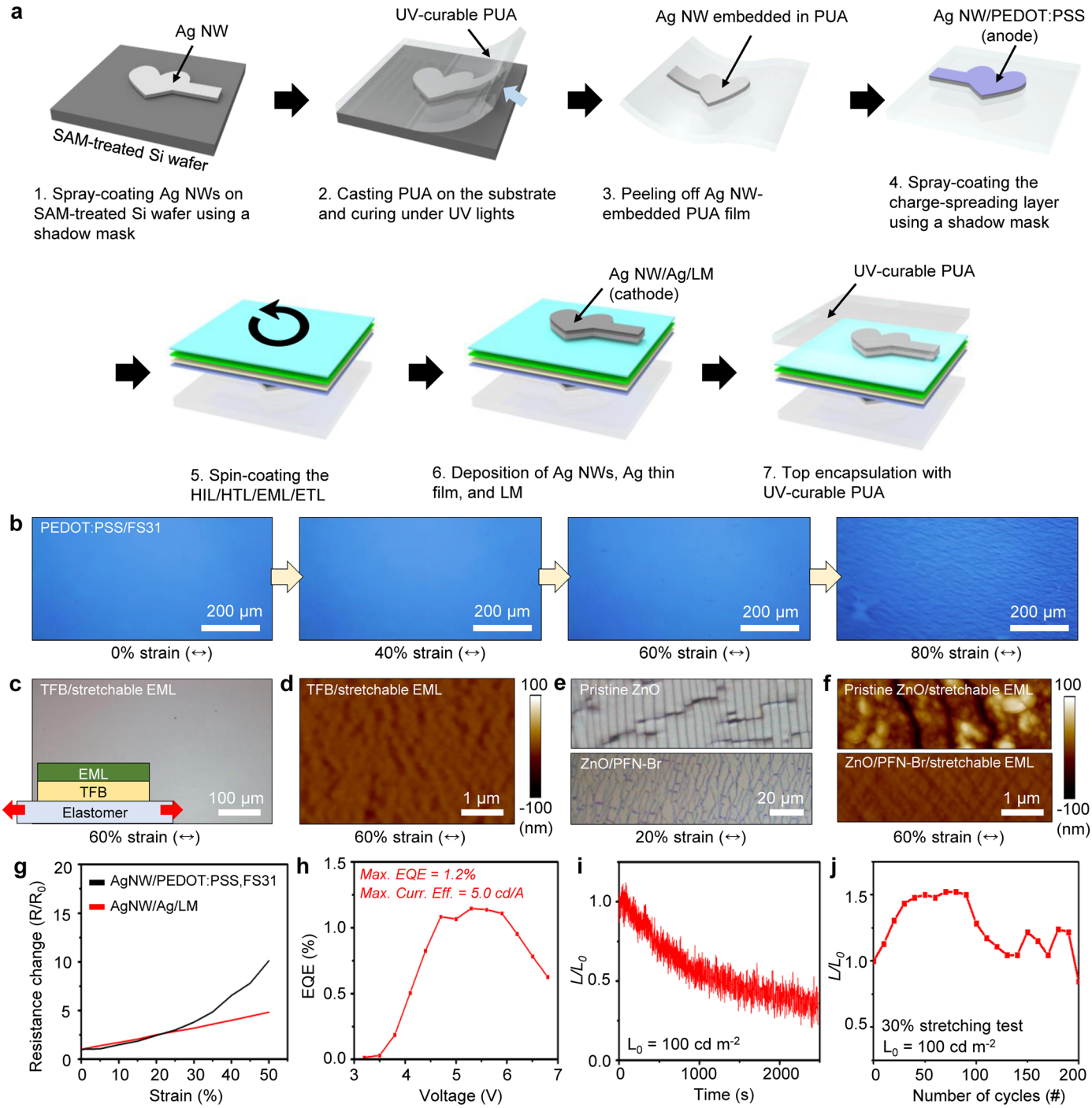

Extended Data Fig. 6 | Fabrication and characterization of intrinsically stretchable QLEDs. a, Illustrations show the step-by-step fabrication process of the intrinsically stretchable QLEDs. **b**, Stretchability of the PEDOT:PSS/FS-31 composite film. **c, d**, Optical (c) and AFM images (d) of the double-layered structure of the TFB/stretchable EML film, stretched under 60% applied strain. The inset schematic image in panel (c) shows the structure of double-layered films. **e**, Optical images of the pristine ZnO film (upper panel) and the ZnO/PFN-Br composite film (lower panel) stretched under 20% applied strain. **f**, AFM images of the double-layered structure of the pristine ZnO/stretchable EML film (upper panel), and the ZnO/PFN-Br/stretchable EML film (lower panel) stretched under 60% applied strain. **g**, Resistance increase of stretchable anodes and cathodes under the applied strain. **h**, EQE of the intrinsically stretchable QLED under increasing voltage. **i**, Device lifetime of the intrinsically stretchable QLED. **j**, Cyclic stretching test data of the intrinsically stretchable QLED.

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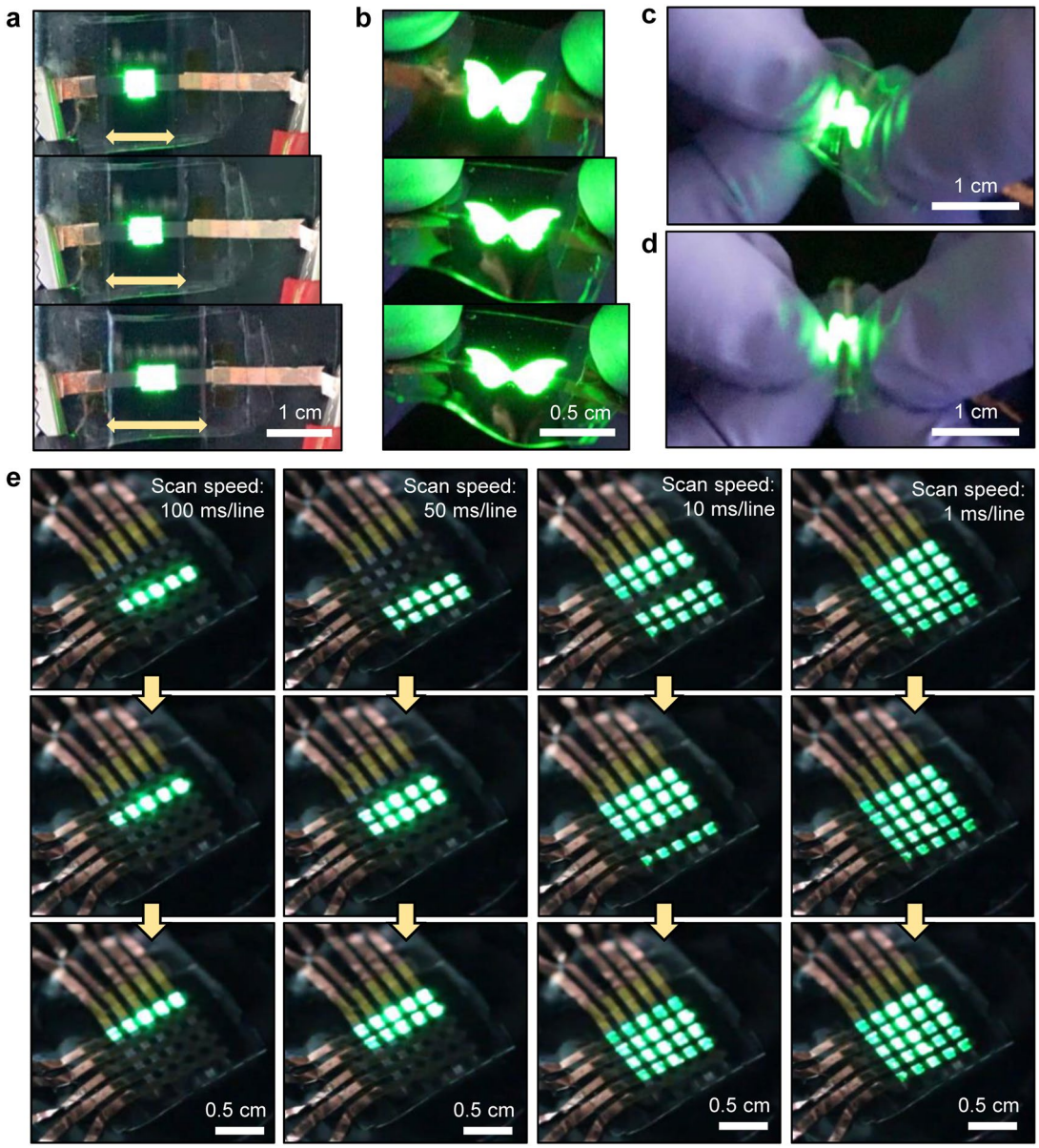

Extended Data Fig. 7 | Green-emitting intrinsically stretchable QLED and PM-QLED array. a-d, Intrinsically stretchable QLEDs under various deformation states. The devices exhibit outstanding mechanical stability under 1-D stretching (a, b), twisting (c), and crumpling (d). **e**, Snapshots of Supplementary Movie 1. Scan speed adjustment of intrinsically stretchable PM-QLED array, operated under common cathode configuration. Scan speed should be faster than 1 ms per line to display a pattern without flickering.

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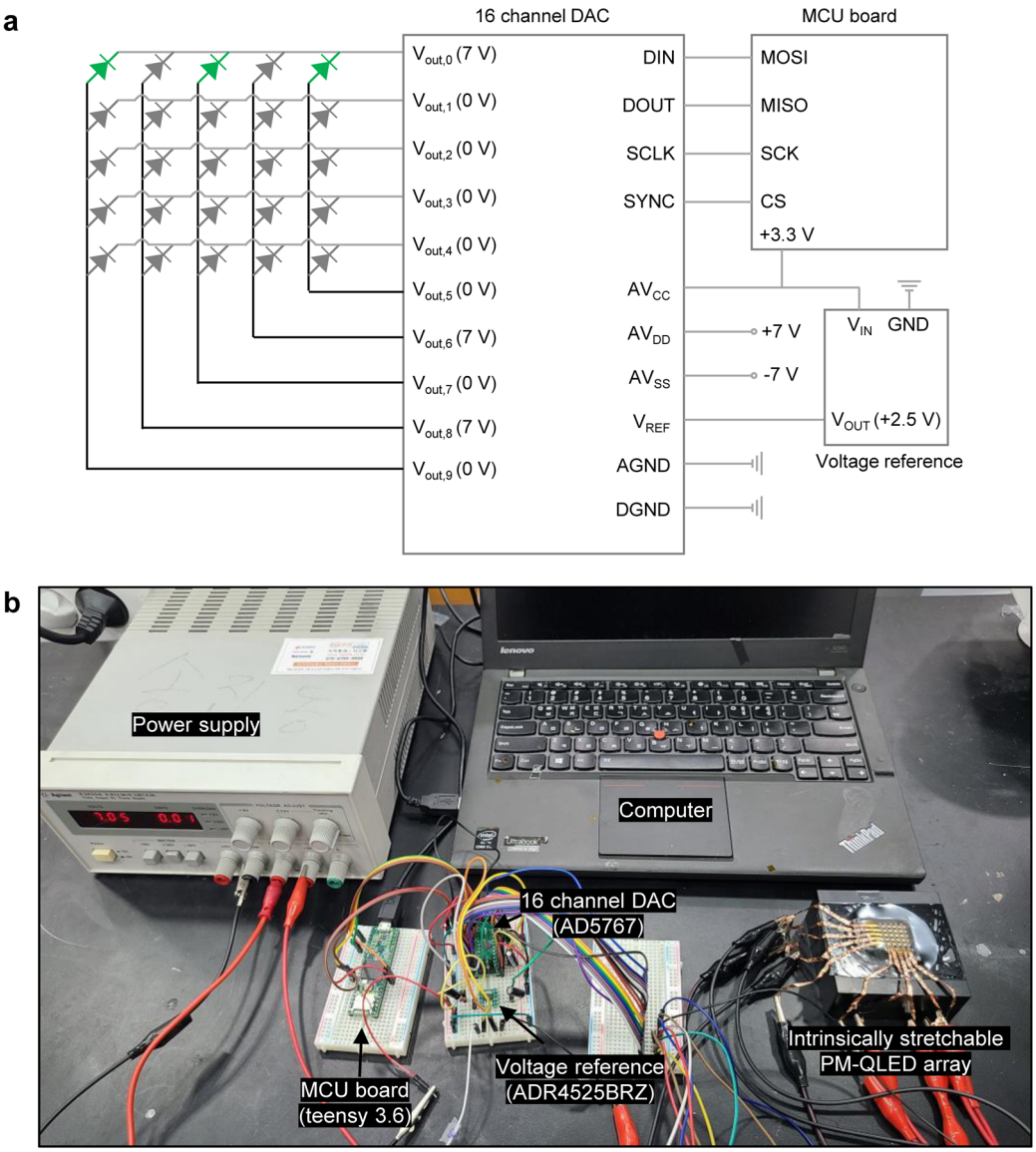

Extended Data Fig. 8 | A customized circuit for the operation of the intrinsically stretchable PM-QLED arrays. a, Circuit diagram of the experimental setup for the operation of the PM-QLED array. The MCU board (teensy 3.6), 16 channel DAC (AD5767), and voltage reference (ADR4525BRZ) were used in the setup. **b**, Photograph of the experimental setup.

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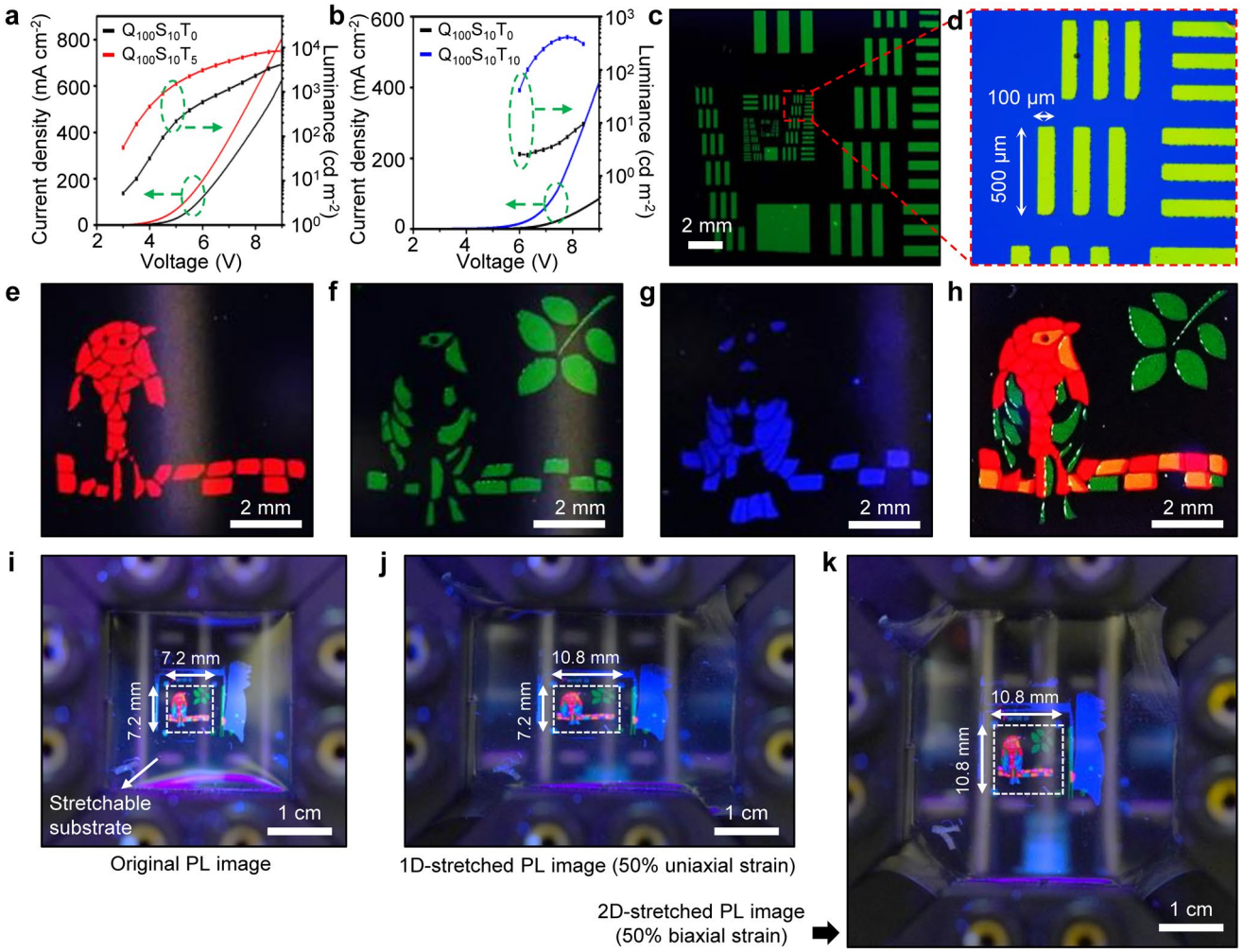

Extended Data Fig. 9 | Aligned multiple transfer printing of the intrinsically stretchable EML patterns. a, Device performance of QLEDs using red-emitting stretchable EMLs. **b**, Device performance of QLEDs using blue-emitting stretchable EMLs. **c, d**, Transfer printing of rectangular patterns of stretchable EML with various pattern sizes (c). The highest resolution of transfer printing is ~100 μm (d). **e-h**, Aligned transfer printing of RGB stretchable EML patterns. Red (e), green (f), blue (g), and red/green (h) stretchable EML patterns are printed in an aligned manner on the stretchable substrate. **i-k**, PL images of the printed stretchable EML patterns. Original images (i), 1D-stretched image under 50% uniaxial strain (j), and 2D-stretched image under 50% biaxial strain (k).

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Extended Data Table 1 | Performance of reported intrinsically stretchable light-emitting devices

Device type	Journal (year)	Stretchable EML		Device structure	Max. luminance (cd m^{-2}, @ V)	Turn-on voltage (V)	V @1000 cd m^{-2} (V)	Max. Stretchability (%)	Ref.
		Emitters	Additives						
Intrinsically stretchable OLEC	*Adv. Mater.* (2011)	PF-B (65 wt%)	PEO-DMA (32 wt%) LiTf (3 wt%)	Anode/ EML/ Cathode	300 (@12 V)	4.8	-	45	10
	Nat. Photon. (2013)	SuperYellow (80 wt%)	ETPTA (8 wt%) PEO (8 wt%) LiTf (4 wt%)	Anode/ EML/ Cathode	2,200 (@21 V)	6.8	21	120	11
	Nat. Commun. (2020)	SuperYellow (46.5 wt%)	Ion-conducting polymer (46.5 wt%) ETT-15 (4.7 wt%) LiTf (2.3 wt%)	Anode/ EML/ Cathode	Max. luminance N/A	15	N/A	30	14
Intrinsically stretchable OLED	*ACS Nano* (2014)	white-emitting polymer (91 wt%)	OXD-7 (9 wt%)	Anode/ HTL/EML/ETL/ Cathode	1,100 (@21 V)	7.0	18	130	47
	Sci. Adv. (2021)	SuperYellow (67 wt%)	Triton-X (33 wt%)	Anode/ HTL/EML/ETL/ Cathode	2,500 (@11 V)	8.3	10	80	15
	Adv. Mater. (2022)	SuperYellow (80 wt%)	Triton-X (20 wt%)	Anode/ HTL/EML/ETL/ Cathode	2,185 (@15 V)	5.3	11	73	16
	Adv. Mater. (2022)	L-SY-PPV (50 wt%)	PAN (50 wt%)	Anode/ HTL/EML/ETL/ Cathode	3,780 (@13 V)	6.5	10	30 (serpentine)	12
	Nature (2022)	SuperYellow (50 wt%)	PU (50 wt%)	Anode/ HTL/EML/ETL/ Cathode	7,450 (@15 V)	5.0	10.5	100	6
	Adv. Funct. Mater. (2023)	SuperYellow (59 vol%)	PEO (35 vol%) KCF_3SO_3 (6 vol%)	Anode/ HIL/EML/ Cathode	1,754 (@11 V)	3.7	7.0	30	13
	Nat. Mater. (2023)	PDKCD (100 wt%)	-	Anode/ HTL/EML/ETL/ Cathode	2,175 (@10 V)	4.7	8.0	60	17
Intrinsically stretchable QLED	**This work**	Colloidal QD (87 wt%)	SEBS-g-MA (8.7 wt%) TFB (4.3 wt%)	Anode/ HIL/HTL/EML/ETL/ Cathode	15,170 (@6.2 V)	3.2	4.3	50	-

Comparison between our work and previously reported d.c.-driven intrinsically stretchable OLECs and OLEDs, in terms of the emitting materials, device structures, maximum luminance, turn-on voltage, voltage at 1000 cd m^{-2} and stretchability.